

Class Work(01.08.26)

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QUESTIONS

Q1 List Comprehension - Reverse
10 marks**Q2** List Comprehension - Sum
10 marks**Q3** List Comprehension - Sum of
ODD
10 marks**Q4** List Comprehension - Sum of
Squares
10 marks**Q5** List Comprehension -Sum of
Numbers Divisible by 3
10 marks**Q1** List Comprehension - Reverse

10 marks

WAP to display in reverse of numbers from 1 to 10 using List Comprehension
output:
[10, 9, 8, 7, 6, 5, 4, 3, 2, 1]

Your Code

```
1 a=[x for x in range(10,0,-1)]  
2 print(a)
```

Input

stdin input

Output

```
[10, 9, 8, 7, 6, 5, 4, 3, 2, 1]
```

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Q2 List Comprehension - Sum

10 marks

WAP to find the Sum of First 10 Natural Numbers
output:
55

Your Code

```
1 a=[x for x in range(1,11)]  
2 tot=sum(a)  
3 print(tot)
```

Input

stdin input

Output

```
The sum of first ten natural  
numbers is 55
```

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Q3 List Comprehension - Sum of ODD

10 marks

WAP to find the Sum of Odd Numbers up to 10
output:
25

Your Code

```
1 a=[x for x in range(1,11,2)]  
2 tot=sum(a)  
3 print(tot)
```

Input

stdin input

Output

25

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Q4 List Comprehension - Sum of Squares

10 marks

WAP to find the Sum of Squares up to 5
output:
55

Your Code

```
1 a=[x**2 for x in range(6)]  
2 tot=sum(a)  
3 print(tot)
```

Input

stdin input

Output

55

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Q5 List Comprehension -Sum of Numbers Divisible by 3

10 marks

WAP to find the Sum of first 12 Numbers Divisible by 3

Output:

30

Your Code

```
1 a=[x for x in range(3,13) if x%3==0]
2 print(sum(a))
```

Input

stdin input

Output

30

Q6 List Comprehension - Factorial

10 marks

WAP to find the factorial of a given number

input:

5

output:

120

Your Code

```
1 import math
2 n=5
3 a = math.prod([x for x in range(1,n+1)])
4 print(a)
```

Input

stdin input

Output

120

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Q7 List Comprehension - Sum of Factorials

10 marks

WAP to find the Sum of Factorials

Series: 1! + 2! + 3! + 4!

Output:

33

Your Code

```
1 import math
2 a=[math.factorial(x) for x in range(1,5)]
3 print(sum(a))
```

Input

stdin input

Output

33

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Q8 List Comprehension - Sum of Digits of a Number

10 marks

WAP to find the Sum of Digits of a Number

input:

4567

output:

22

Your Code

```
1 n=4567
2 a=[int(x) for x in str(n)]
3 print(sum(a))
```

Input

stdin input

Output

22

Q9 List Comprehension - Sum of Positive Numbers in a List

10 marks

WAP to find the Sum of Positive Numbers in a List

input:

[10,-5,20,-8,15,-2]

output:

45

Your Code

```
1 a=[10,-5,20,-8,15,-2]
2 n=[x for x in a if x>0]
3 print(sum(n))
```

Input

stdin input

Output

45

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Q10 List Comprehension - Sum of Squares of Even Numbers

10 marks

WAP to Find the Sum of the Squares of Even Numbers up to 10

$(2^2 + 4^2 + 6^2 + 8^2 + 10^2)$

OUTPUT:

220

Your Code

```
1 a=[x**2 for x in range(2,11,2)]
2 print(sum(a))
```

Input

stdin input

Output

220

Visible Test Cases